

PREPARATION SHEET LAB 03

FIR FILTER IMPLEMENTATION IN MATLAB AND IN C

Team (Group number + letter as chosen in Moodle)	Team members	
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	3	

Please fill out the document electronically – including the yellow fields above – and upload it to Moodle once per team by the start of the lab event (no matter who uploads it, it will automatically be assigned to all members). Please note that the complete assignment can be found in another document.

Prep task (for short test)

Familiarize yourself with the concepts of

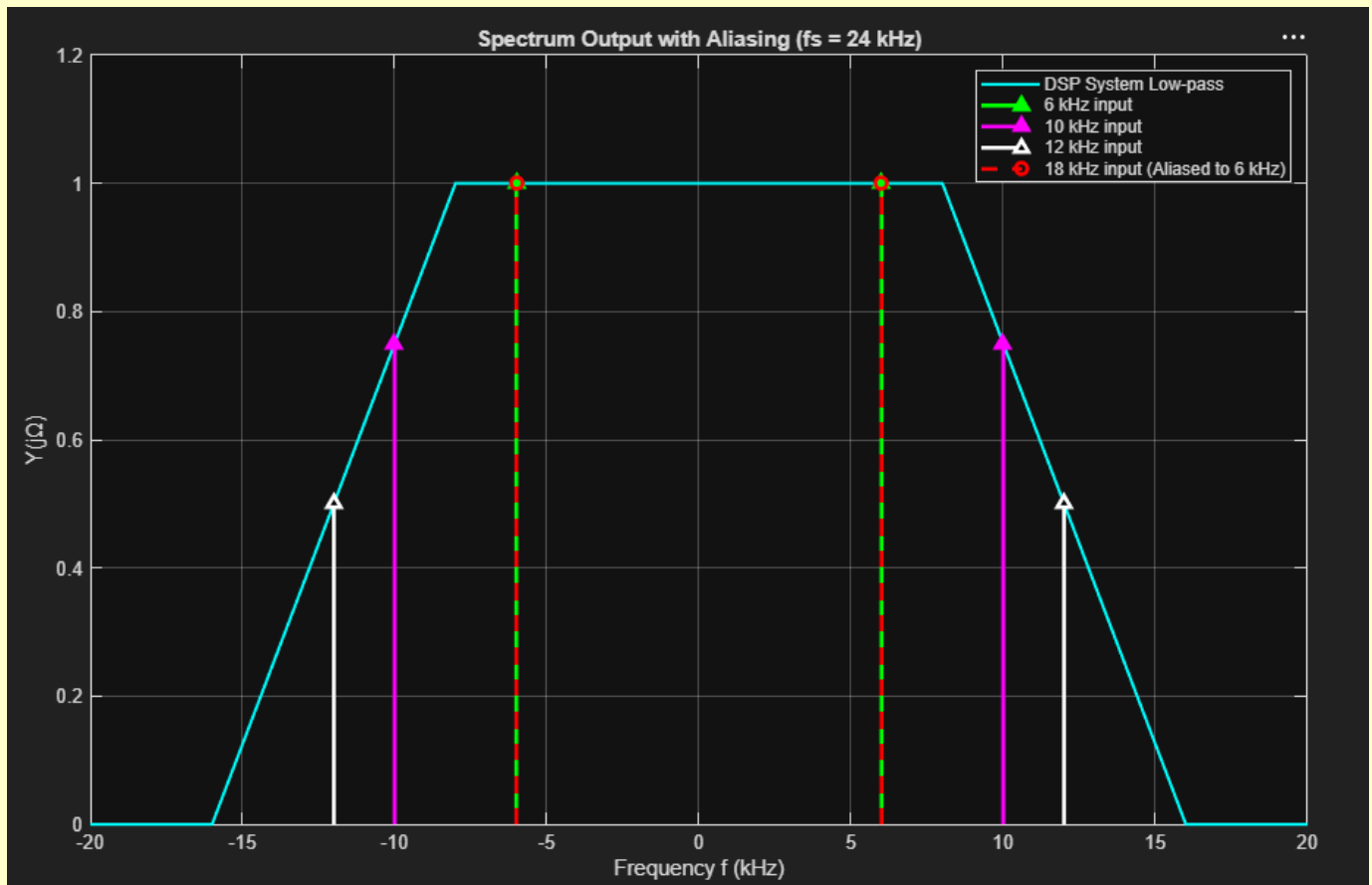
- Finite Impulse Response (FIR) filters, including
- description of FIR filters by impulse response, transfer function, frequency response, pole zero plot and
- properties like symmetries, linear-phase and group delay.

These topics will be addressed by the short test at the beginning of the lab session.

Prep task 1: Sampling frequency and aliasing

- Explain for what purpose the DSP system has a low-pass filter characteristic.
- Sketch the spectrum $Y(j\Omega)$ of the analog signal $y(t)$ at the output of the DSP system

- Anti-Aliasing Filter (before ADC): It restricts the bandwidth of the incoming analog signal to below the Nyquist frequency ($f_s / 2$). If frequencies higher than this enter the ADC, they alias into the baseband and irreversibly corrupt the digital signal.
- Reconstruction Filter (after DAC): Also known as an anti-imaging filter, it smooths out the "staircase" output from the digital-to-analog conversion by filtering out the high-frequency spectral images created by the zero-order hold process.



Prep task 2: lowpass/bandstop filter design

1. Write a MATLAB script *FIR_lab.m* that designs a lowpass filter with the above specification. Use the MATLAB functions *firpmord()* and *firpm()*.

- Give the Matlab code here:

% 1. Specifications for Lowpass Filter

fs = 25000; % Sampling frequency in Hz (fs)

f_pass = 4500; % Passband edge frequency in Hz (where the filter starts cutting)

f_stop = 5000; % Stopband edge frequency in Hz (where the filter must reach maximum attenuation)

Rp_lin = 0.1; % Maximum passband ripple (given as a linear value, not in decibels)

As_dB = 40; % Minimum stopband attenuation (given in decibels)

dev_pass = Rp_lin; % Already linear, so we just pass it through

dev_stop = 10^(-As_dB / 20); % Formula to convert dB to linear magnitude

```

% 2. Design the Lowpass filter

f_edges = [f_pass f_stop]; % The transition band boundaries

amps = [1 0]; % Desired amplitude at those bands (1 = pass, 0 = stop)

devs = [dev_pass dev_stop]; % Maximum allowed error in those bands

[n, fo, ao, w] = firlmord(f_edges, amps, devs, fs); % Minimum filter order (n), normalized frequency (fo),
amplitude (ao), weight (w).

if mod(n, 2) ~= 0

    n = n + 1; % % If 'n' is odd, add 1 to force it to be even

end

b_fir_coe_LP = firpm(n, fo, ao, w); % optimization algorithm) to calculate the actual impulse response
(coefficients).

% 3. Bandstop Filter via Frequency Transformation

b_fir_coe_BS = zeros(1, 2 * length(b_fir_coe_LP) - 1); % original length, minus 1

b_fir_coe_BS(1:2:end) = b_fir_coe_LP;

```

2. Determine the coefficients of an bandstop filter from the designed lowpass filter and report the coefficients of both filters.
3. Plot the impulse responses of the lowpass filter *b_fir_coe_LP* and the bandstop filter *b_fir_coe_BS*.

- Coefficients, impulse responses

LP Filter Coefficients->

Columns 1 through 10:

-0.0086 -0.0040 0.0065 0.0193 0.0229 0.0129 -0.0020 -0.0076 0.0004 0.0108

Columns 11 through 20:

0.0094 -0.0037 -0.0129 -0.0058 0.0103 0.0152 0.0005 -0.0172 -0.0149 0.0081

Columns 21 through 30:

0.0246 0.0109 -0.0210 -0.0315 -0.0005 0.0407 0.0372 -0.0236 -0.0789 -0.0409

Columns 31 through 40:

0.1119 0.2941 0.3755 0.2941 0.1119 -0.0409 -0.0789 -0.0236 0.0372 0.0407

Columns 41 through 50:

-0.0005 -0.0315 -0.0210 0.0109 0.0246 0.0081 -0.0149 -0.0172 0.0005 0.0152

Columns 51 through 60:

0.0103 -0.0058 -0.0129 -0.0037 0.0094 0.0108 0.0004 -0.0076 -0.0020 0.0129

Columns 61 through 65:

0.0229 0.0193 0.0065 -0.0040 -0.0086

Band stop Filter Coefficients:

Columns 1 through 10

-0.0086 0 -0.0040 0 0.0065 0 0.0193 0 0.0229 0

Columns 11 through 20

0.0129 0 -0.0020 0 -0.0076 0 0.0004 0 0.0108 0

Columns 21 through 30

0.0094 0 -0.0037 0 -0.0129 0 -0.0058 0 0.0103 0

Columns 31 through 40

0.0152 0 0.0005 0 -0.0172 0 -0.0149 0 0.0081 0

Columns 41 through 50

0.0246 0 0.0109 0 -0.0210 0 -0.0315 0 -0.0005 0

Columns 51 through 60:

0.0407 0 0.0372 0 -0.0236 0 -0.0789 0 -0.0409 0

Columns 61 through 70:

0.1119 0 0.2941 0 0.3755 0 0.2941 0 0.1119 0

Columns 71 through 80:

-0.0409 0 -0.0789 0 -0.0236 0 0.0372 0 0.0407 0

Columns 81 through 90:

-0.0005 0 -0.0315 0 -0.0210 0 0.0109 0 0.0246 0

Columns 91 through 100:

0.0081 0 -0.0149 0 -0.0172 0 0.0005 0 0.0152 0

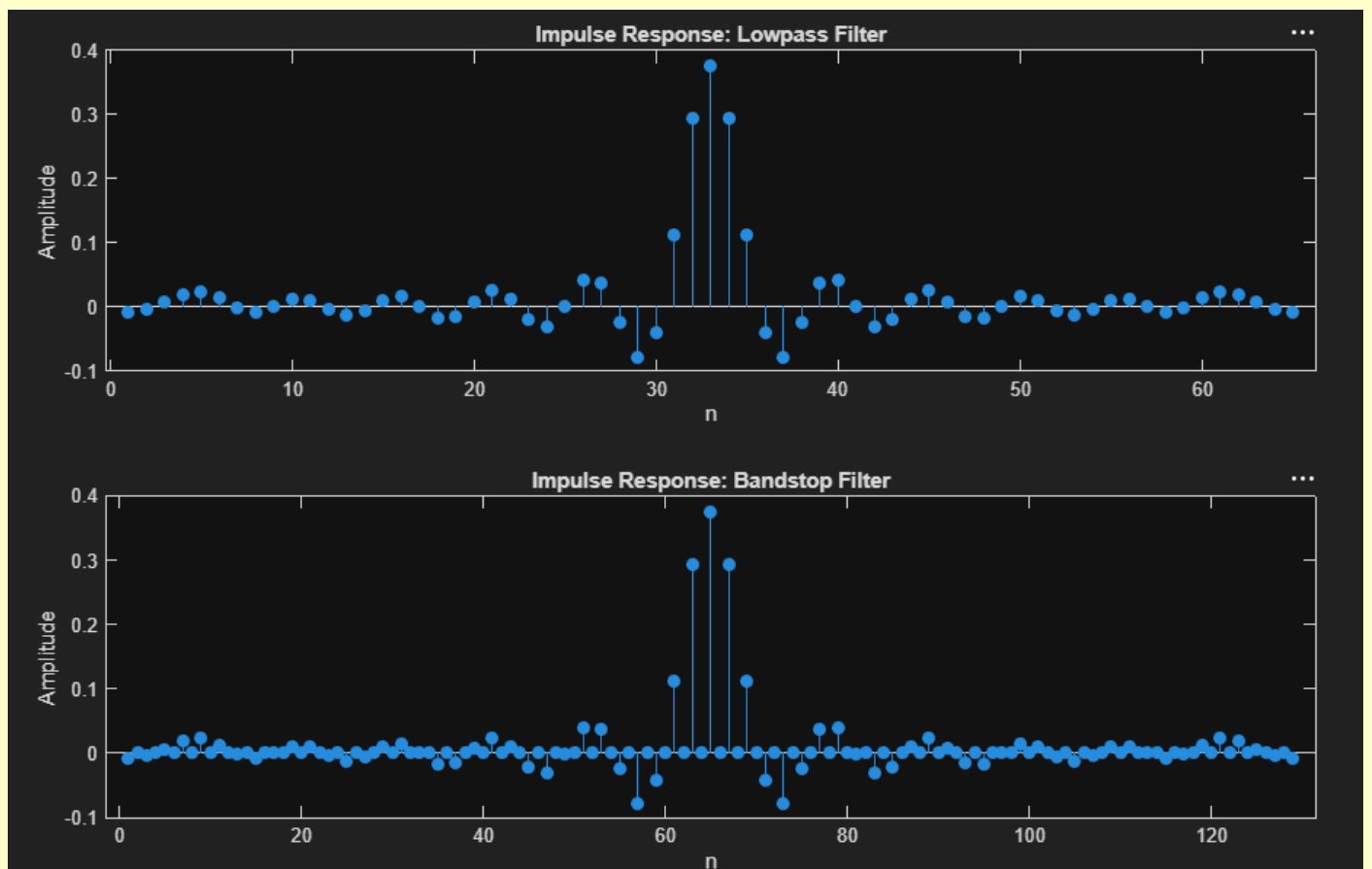
Columns 101 through 110: 0.0103 0 -0.0058 0 -0.0129 0 -0.0037 0

0.0094 0

Columns 111 through 120:

0.0108 0 0.0004 0 -0.0076 0 -0.0020 0 0.0129 0

Columns 121 through 129: 0.0229 0 0.0193 0 0.0065 0 -0.0040 0 -
0.0086



4. Specify the four edge frequencies of the bandstop filter. These result from the edge frequencies of the lowpass filter.
5. Plot the amplitude responses in dB with coefficients corresponding to the precision of *int16_t* (16 bits). Use "round" in combination with an appropriate scaling factor.
6. Check all parameters of the specification. If it is not complied with, the filter order may have to be increased. This would have to be done after calling *firpmord*().

- Edge frequencies (part 4)

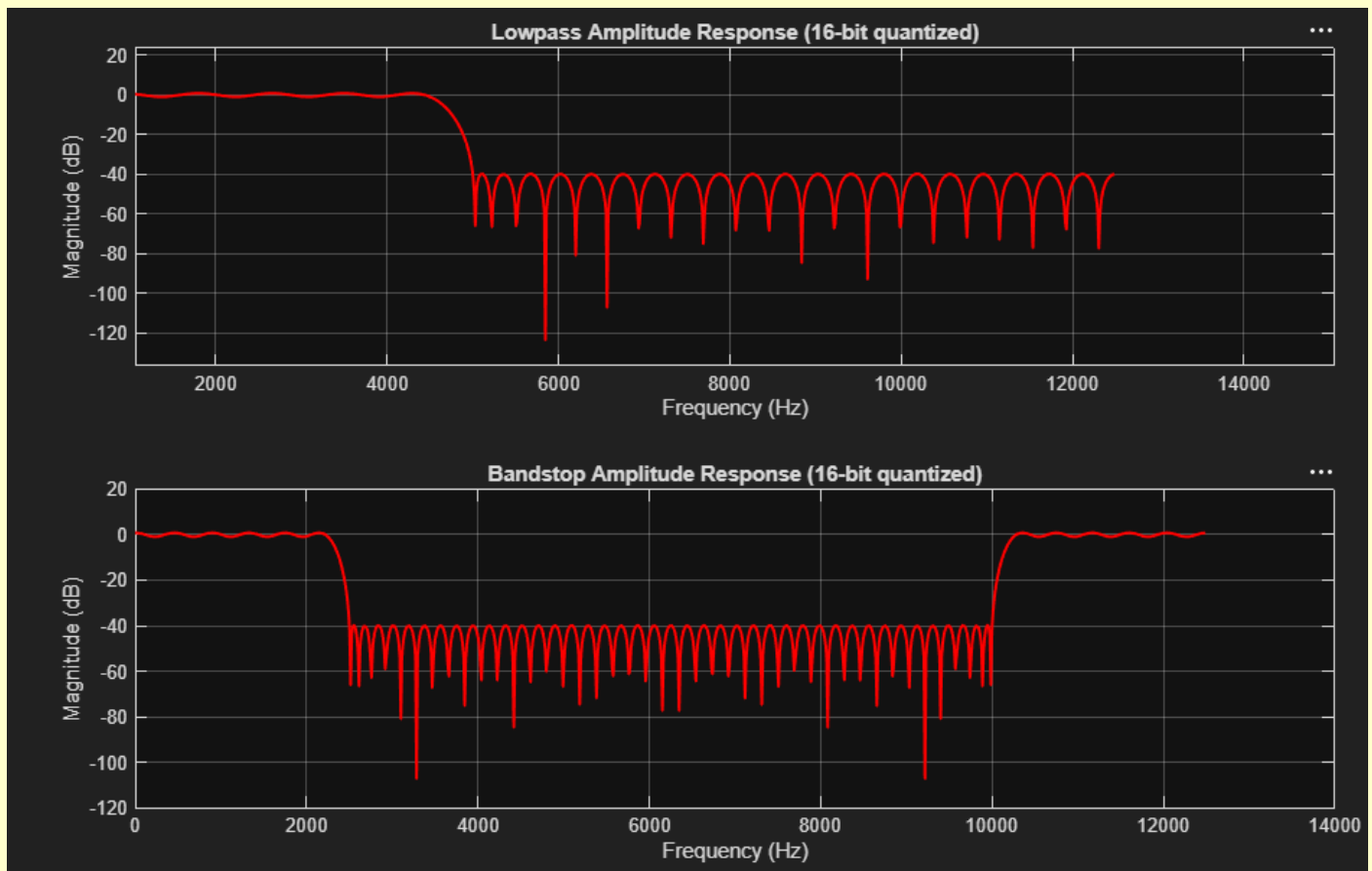
Lower passband edge: $4500 / 2 = 2250$ Hz

Lower stopband edge: $5000 / 2 = 2500$ Hz

Upper stopband edge: $12500 - 2500 = 10000$ Hz

Upper passband edge: $12500 - 2250 = 10250$ Hz

- Plot the amplitude responses (using dB-scale) with coefficients quantized to 16 bit.



- Check if corner frequencies of the filter, pass band ripple and stop band attenuation are within specification limits. If not met, increase the FIR filter order until the specification is met.

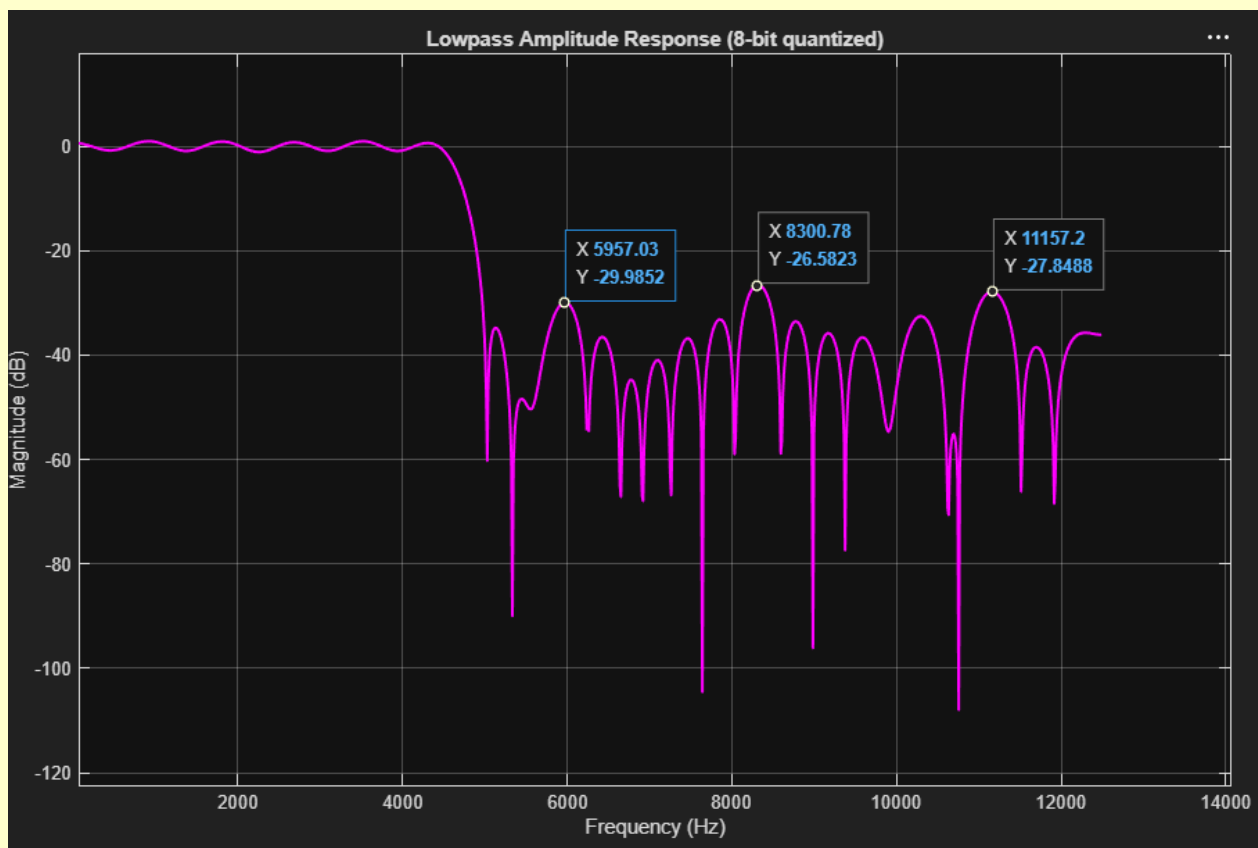
All parameters of the specification are met with the 16-bit quantized coefficients.

7. Create *int16_t* (16 bits) filter coefficients and write each to a header file, *FIR_LP.h* resp. *FIR_BS.h*. Use the function *write_coeffs.m* for this, which is provided in Moodle. Use the code lines for calling the function at the end of Listing 3.1.
8. Quantize the coefficients of the lowpass filter to 8 bit precision. Plot the amplitude responses in dB. What attenuation do you achieve with the quantized coefficients in the stopband?

- Content of header files.

```
const int16_t asFirCoeLP[65] = {-283, -130, 212, 631, 752, 423, -67, -250, 13, 354, 307, -121,
-422, -189, 338, 497, 18, -565, -487, 266, 806, 356, -688, -1032, -17, 1335, 1218, -772, -2586,
-1341, 3667, 9637, 12305, 9637, 3667, -1341, -2586, -772, 1218, 1335, -17, -1032, -688, 356,
806, 266, -487, -565, 18, 497, 338, -189, -422, -121, 307, 354, 13, -250, -67, 423, 752, 631,
212, -130, -283};
```

- Measured attenuation in stop band: ~26dB



Prep task 3: Highpass

Extend your MATLAB script *FIR_lab.m* to generate the coefficients for a highpass from the lowpass coefficients.

1. Generate the coefficients of an all-pass filter that has the same order as the lowpass filter. Use the example discussed in the lecture.
2. Subtract the lowpass coefficients from the allpass coefficients.
3. Plot the amplitude response of the resulting highpass filter in dB.
4. Give the full specification of the high-pass filter and explain how this is achieved.
5. Create *short int* filter coefficients and write them in a header file *FIR_HP.h*.

- MATLAB code for part 1&2:

```
% 1. Generate All-pass coefficients (Delay M)
```

```
M = n / 2; % Center delay index
```

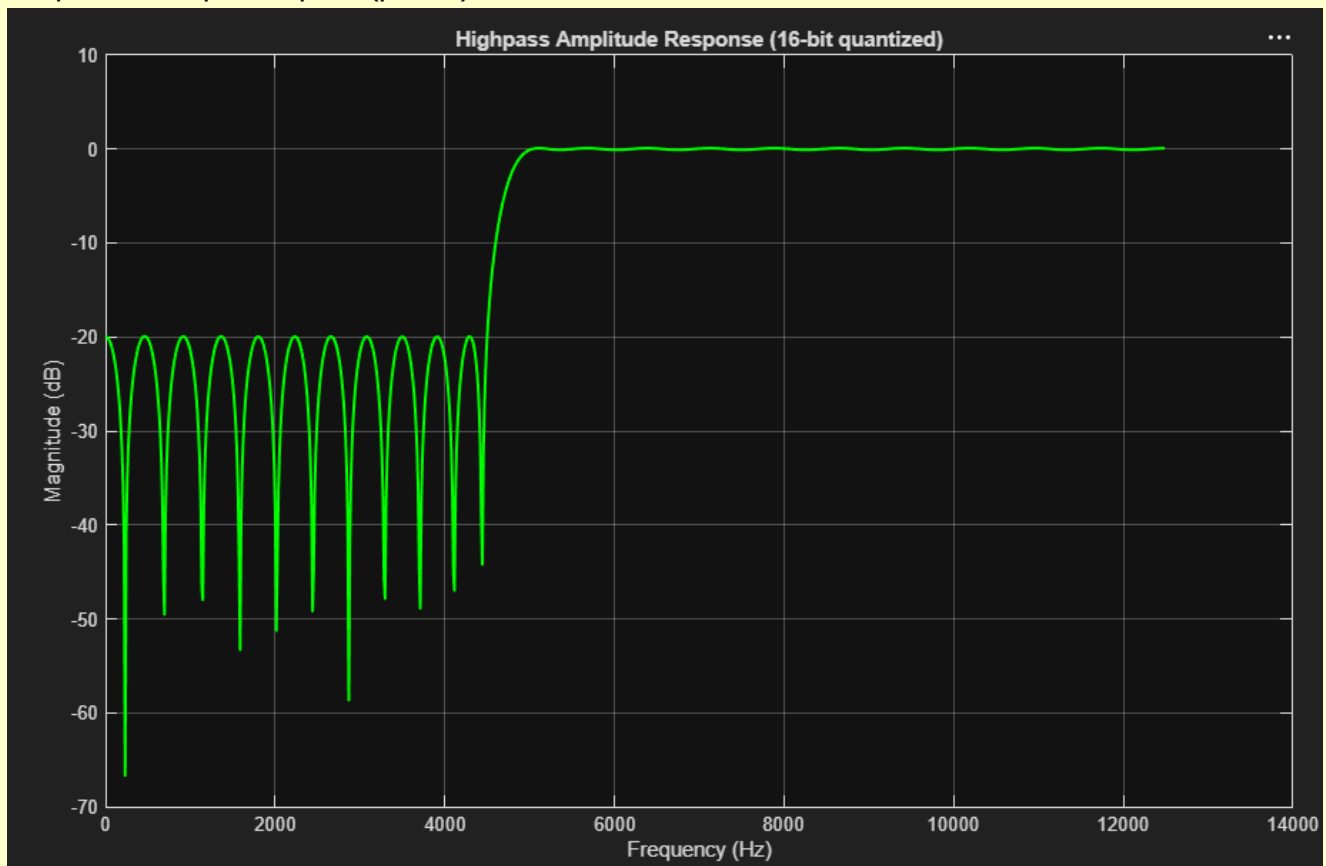
```
allpass = zeros(1, length(b_fir_coe_LP));
```

```
allpass(M + 1) = 1;
```

```
% 2. Subtract lowpass from allpass to create Highpass
```

```
b_fir_coe_HP = allpass - b_fir_coe_LP;
```

Amplitude response plot (part 3)



- Spec high pass filter (part 4):

The highpass filter is generated by spectral inversion:

$$H_{HP}(z) = z^{(-M)} - H_{LP}(z)$$

The stopband edge of the highpass is 4500 Hz.

The passband edge of the highpass is 5000 Hz

The highpass stopband attenuation is determined by the lowpass passband ripple:

$$A_{s,HP} = -20 \log_{10}(0.1) = 20 \text{ dB}$$

The highpass passband ripple is determined by the lowpass stopband error:

$$\delta_{p,HP} = 10^{(-40/20)} = 0.01$$

Therefore, the resulting highpass has approximately:

- stopband edge: 4500 Hz
- passband edge: 5000 Hz
- stopband attenuation: 20 dB
- passband ripple: 0.01 linear, approximately ± 0.087 dB

- Content of header file (part 5):
- ```
const int16_t asFirCoeHP[65] = {283, 130, -212, -631, -752, -423, 67, 250, -13, -354, -307, 121, 422, 189, -338, -497, -18, 565, 487, -266, -806, -356, 688, 1032, 17, -1335, -1218, 772, 2586, 1341, -3667, -9637, 20462, -9637, -3667, 1341, 2586, 772, -1218, -1335, 17, 1032, 688, -356, -806, -266, 487, 565, -18, -497, -338, 189, 422, 121, -307, -354, -13, 250, 67, -423, -752, -631, -212, 130, 283};
```